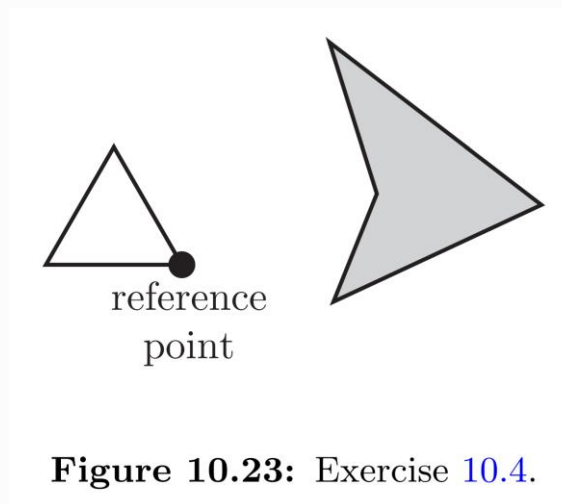


Foundations of Robotics

Homework 8

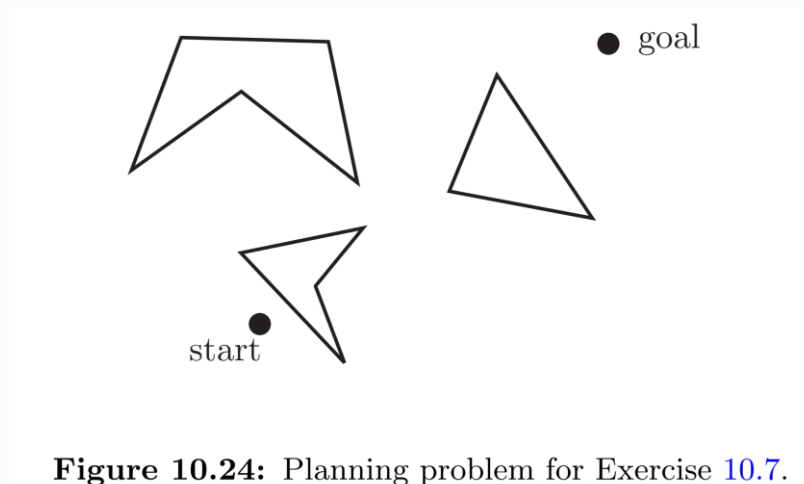
1. (Exercise 10.4) Please solve the exercise introduced as follows.

Exercise 10.4 Draw the C-obstacle corresponding to the obstacle and translating planar robot in Figure 10.23.



2. (Exercise 10.7) Please solve the exercise introduced as follows.

Exercise 10.7 Draw the visibility roadmap for the C-obstacles and q_{start} and q_{goal} in Figure 10.24. Indicate the shortest path.



3. The translating-only triangle-shape robot shown in the figure below (origin at the bottom right corner) has to reach the goal position marked with a star. The space is wide open, meaning the end of the grid is not a wall. Sketch the obstacle configuration space. Does a path to the goal exist? If so, draw one feasible path.

